

CHAPTER 15

The Main Conjecture and Annihilation of Class Groups

In the mid 1980s, Thaine and Kolyvagin invented new techniques for constructing relations in ideal class groups. These methods have had profound consequences. Not only is it now possible to give a fairly elementary proof of the Main Conjecture, but these ideas also allowed Rubin to give the first examples of finite Tate–Shafarevich groups for elliptic curves.

In the following, we prove four main theorems, each one requiring an extension of the ideas needed for the previous one. We start in Section 15.1 by giving an easy proof of Stickelberger’s theorem. In Section 15.2, we prove a result of Thaine on annihilation of ideal classes of real abelian fields. In Section 15.3, we introduce an iterative procedure that is due to Kolyvagin and which has Thaine’s method as its first step. As a consequence, we obtain Ribet’s converse of Herbrand’s theorem. In the last four sections, we use Kolyvagin’s method to prove the Main Conjecture for $\mathbb{Q}(\zeta_p)$, following Rubin [7].

§15.1. Stickelberger’s Theorem

In the following sections we will obtain annihilators of ideal class groups by various techniques. The basic theorem along this line is Stickelberger’s theorem (Theorem 6.10), and the present methods can be used to give an easy proof of this result, at least for the case of full cyclotomic fields. A hint of the present proof appears in Kummer [6, p. 220].

Let m be a positive integer and let $G = \text{Gal}(\mathbb{Q}(\zeta_m)/\mathbb{Q}) \simeq (\mathbb{Z}/m\mathbb{Z})^\times$.

Theorem 15.1. *Let*

$$\theta = \frac{1}{m} \sum_{\substack{a=1 \\ (a,m)=1}}^m a\sigma_a^{-1}.$$

Let $\beta \in \mathbb{Z}[G]$ be such that $\beta\theta \in \mathbb{Z}[G]$. Then $\beta\theta$ annihilates the ideal class group of $\mathbb{Q}(\zeta_m)$.

Proof. Let $\mathfrak{C}$ be an ideal class of $\mathbb{Q}(\zeta_m)$. There exist infinitely many unramified prime ideals of degree 1 in $\mathfrak{C}$. Choose such a prime ideal λ and let ℓ be the rational prime below λ . Since λ is of degree 1, ℓ splits completely in $\mathbb{Q}(\zeta_m)$, so $\ell \equiv 1 \pmod{m}$. Fix a primitive root $s \pmod{\ell}$. Define a character $\chi: (\mathbb{Z}/\ell\mathbb{Z})^\times \rightarrow \mathbb{C}^\times$ of order m by setting $\chi(s) = \zeta_m$. Let

$$g(\chi) = - \sum_{b=1}^{\ell-1} \chi(b) \zeta_\ell^b$$

be a Gauss sum. Let $\mathcal{L}$ be the prime of $\mathbb{Q}(\zeta_m, \zeta_\ell)$ above λ and let

$$v_{\sigma_a^{-1}\mathcal{L}}(g(\chi)) = r_a,$$

where σ_a is extended to $\mathbb{Q}(\zeta_m, \zeta_\ell)$. Since $g(\chi)\overline{g(\chi)} = \ell$, we have $0 \leq r_a \leq \ell - 1$. By Lemma 6.4, $g(\chi)^{\ell-1} \in \mathbb{Q}(\zeta_m)$. Since $\mathcal{L}^{\ell-1} = \lambda$,

$$v_{\sigma_a^{-1}\lambda}(g(\chi)^{\ell-1}) = r_a.$$

Since only primes above ℓ appear in the factorization of $g(\chi)$, we obtain

$$(g(\chi)^{\ell-1}) = \prod_{(a,m)=1} \sigma_a^{-1} \lambda^{r_a},$$

which says that $\sum r_a \sigma_a^{-1}$ annihilates the class $\mathfrak{C}$ of λ in the class group of $\mathbb{Q}(\zeta_m)$.

Define $\tau \in \text{Gal}(\mathbb{Q}(\zeta_m, \zeta_\ell)/\mathbb{Q}(\zeta_m))$ by $\tau: \zeta_\ell \mapsto \zeta_\ell^s$. Then τ is in the inertia group for $\sigma_a^{-1}\mathcal{L}$, hence acts trivially mod $\sigma_a^{-1}\mathcal{L}$. An easy calculation shows that $g(\chi)^\tau = \chi(s)^{-1}g(\chi)$ and $(\zeta_\ell^s - 1)/(\zeta_\ell - 1) \equiv 1 + \zeta_\ell + \cdots + \zeta_\ell^{s-1} \equiv s \pmod{\sigma_a^{-1}\mathcal{L}}$. Therefore

$$\frac{g(\chi)}{(\zeta_\ell - 1)^{r_a}} \equiv \frac{g(\chi)^\tau}{(\zeta_\ell^s - 1)^{r_a}} \equiv \frac{g(\chi)}{(\zeta_\ell - 1)^{r_a}} \frac{\chi(s)^{-1}}{s^{r_a}} \pmod{\sigma_a^{-1}\mathcal{L}}.$$

Since $v_{\sigma_a^{-1}\mathcal{L}}(\zeta_\ell - 1) = 1$, we have $g(\chi)/(\zeta_\ell - 1)^{r_a}$ relatively prime to $\sigma_a^{-1}\mathcal{L}$. Therefore

$$\zeta_m = \chi(s) \equiv s^{-r_a} \pmod{\sigma_a^{-1}\mathcal{L}}.$$

Note that both sides of this last congruence are in $\mathbb{Q}(\zeta_m)$, so the congruence holds mod $\sigma^{-1}\lambda$. Apply σ_a to obtain

$$\zeta_m^a \equiv s^{-r_a} \pmod{\lambda}.$$

Since the m th roots of unity are distinct mod λ , the order of $\zeta_m \pmod{\lambda}$ is exactly m , so we have

$$\zeta_m \equiv s^{-(\ell-1)c/m} \pmod{\lambda}$$

with $(c, m) = 1$. Therefore

$$r_a \equiv \frac{(\ell - 1)ac}{m} \pmod{\ell - 1}.$$

This congruence implies $r_a \not\equiv 0 \pmod{\ell - 1}$. Since $0 \leq r_a \leq \ell - 1$, we have

$$r_a = (\ell - 1) \left\{ \frac{ac}{m} \right\},$$

where $\{\cdot\}$ denotes the fractional part. Therefore

$$\sum_{(a, m)=1} (\ell - 1) \left\{ \frac{ac}{m} \right\} \sigma_a^{-1} = (\ell - 1) \sigma_c \theta$$

annihilates $\mathfrak{C}$:

$$\lambda^{(\ell-1)\sigma_c\theta} = g(\chi)^{\ell-1}.$$

Now let $\beta \in \mathbb{Z}[G]$ be such that $\beta\theta \in \mathbb{Z}[G]$. Let $\gamma = g(\chi)^{\sigma_c^{-1}\beta}$, so $\gamma^{\ell-1} \in \mathbb{Q}(\zeta_m)$ and $\lambda^{\beta\theta(\ell-1)} = (\gamma^{\ell-1})$. Therefore $\gamma^{\ell-1}$ is the $(\ell - 1)$ st power of an ideal in $\mathbb{Q}(\zeta_m)$, so the extension $\mathbb{Q}(\zeta_m, \gamma)/\mathbb{Q}(\zeta_m)$ can only be ramified at the primes dividing $\ell - 1$ (Exercise 9.1). But

$$\mathbb{Q}(\zeta_m) \subseteq \mathbb{Q}(\zeta_m, \gamma) \subseteq \mathbb{Q}(\zeta_m, \zeta_\ell),$$

so $\mathbb{Q}(\zeta_m, \gamma)/\mathbb{Q}(\zeta_m)$ is totally ramified at the primes above ℓ . It follows that this extension is trivial and $\gamma \in \mathbb{Q}(\zeta_m)$. Therefore we may take the $(\ell - 1)$ st root of the above relation and obtain

$$\lambda^{\beta\theta} = (\gamma)$$

as ideals of $\mathbb{Q}(\zeta_m)$. Therefore $\mathfrak{C}^{\beta\theta} = 1$, as desired. $\square$

§15.2. Thaine's Theorem

Stickelberger's theorem gives annihilators of the minus part of the class group, but gives no useful information for totally real fields. Thaine [3] showed how to use cyclotomic units to obtain analogues of Stickelberger elements for real abelian fields. The method has had applications beyond cyclotomic fields; for example, Rubin [2], [3] used Thaine's method, applied to elliptic units, in his proof that the Tate–Shafarevich groups of certain elliptic curves are finite.

An important ingredient in the proof is what is known as Hilbert's Theorem 90. It is interesting to note that this result was known to Kummer, and the first application of it appears to be in Kummer [6, Section II], where he gives an argument very similar to the first part of the proof below. However, he did not apply the technique to annihilate class groups, but rather to study reciprocity laws. The argument seems to have been overlooked by everyone